

DIGITAL SIGNAL PROCESSING

LABORATORIUM MODULES



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Chapter 1

Introduction to MATLAB (*Pt.1*)

1.1 Objectives

After a practicum assignment, it is expected that:

1. Recognize the environment in the MATLAB program
2. Students can understand some of the main functions that exist in MATLAB

1.2 Theory

The name MATLAB stands for MATrix LABoratory. MATLAB was written originally to provide easy access to matrix software developed by the LINPACK (linear system package) and EISPACK (Eigen system package) projects. MATLAB [1] is a high-performance language for technical computing. It integrates computation, visualization, and programming environment. Furthermore, MATLAB is a modern programming language environment: it has sophisticated data structures, contains built-in editing and debugging tools, and supports object-oriented programming. These factors make MATLAB an excellent tool for teaching and research. It has powerful built-in routines that enable a very wide variety of computations. It also has easy to use graphics commands that make the visualization of results immediately available. Specific applications are collected in packages referred to as toolbox. There are toolboxes for signal processing, symbolic computation, control theory, simulation, optimization, and several other fields of applied science and engineering.

1.3 Starting Matlab

When you start MATLAB, a special window called the MATLAB desktop appears. The desktop is a window that contains other windows. The major tools within or accessible from the desktop are:

1. The Command Window
2. The Command History
3. The Workspace
4. The Current Directory
5. The Help Browser

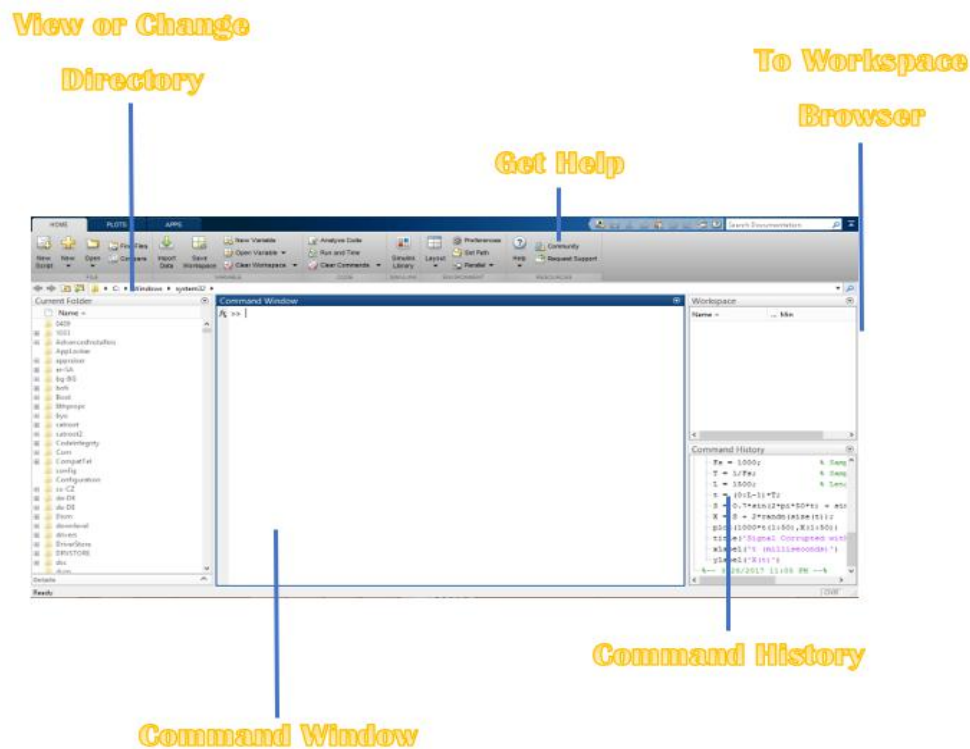


Figure 1.1: Starting Matlab

1.4 Creating Simple Plots *(Cont.)*

The basic MATLAB graphing procedure, for example in 2D, is to take a vector of x-coordinates, $x = (x_1, \dots, x_N)$, and a vector of y-coordinates, $y = (y_1, \dots, y_N)$, locate the points (x_i, y_i) , with $i = 1, 2, \dots, n$ and then join them by straight lines. You need to prepare x and y in an identical array form; namely, x and y are both row arrays or column arrays of the same length. The MATLAB command to plot a graph is `plot(x,y)`.

The vectors $x = (1, 2, 3, 4, 5, 6)$ and $y = (3, -1, 2, 4, 5, 1)$.

The MATLAB command to plot the Vectors is;

```

1      x = [1 2 3 4 5 6];
2      y = [3 -1 2 4 5 1];
3      plot (x, y)

```

The command will show a Signal Chart like figure 1.2.

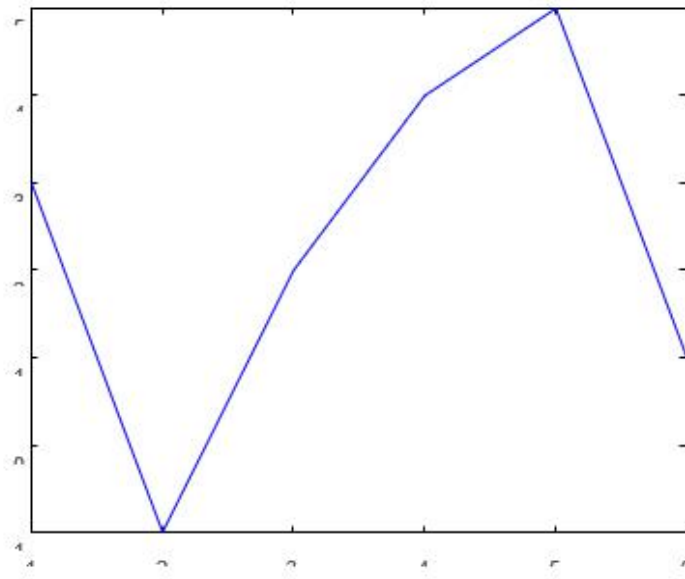


Figure 1.2: Signal Chart

1.5 Assignment

1. Have you understand the basic Function of MATLAB Program(yes/no)?
2. Make your Conclusion about MATLAB Program!
3. Put the picture of your first matlab code and give a brief explanation of the picture !

1.6 Notes

Please write down your written task using Microsoft Word (format: DSP_LAB1_studentID.doc or docx) and the matlab code with the same name (format: DSP_LAB1_studentID.m). The tasks must be finished at lab (no take home lab task), late submission of works will not be marked.

Chapter 2

Introduction to MATLAB (*Pt.2*)

2.1 Objectives

After a practicum assignment, it is expected that:

1. Students can Understand the meaning of Sine and Cosine Wave.
2. Students can implement the Sine and Cosine formula into MATLAB Command.
3. Students can Represent the output Signal of Sine and Cosine Wave using MATLAB Program.

2.2 Theory

2.2.1 Sine Wave

A sine wave or sinusoid is a mathematical curve that describes a smooth repetitive oscillation. A sine wave is a continuous wave. It is named after the function sine, of which it is the graph. It occurs often in pure and applied mathematics, as well as physics, engineering, signal processing and many other fields. Its most basic form as a function of time (t) is:

$$y(t) = A\sin(2\pi ft + \varphi) = A\sin(\omega t + \varphi)$$

where

- A = the amplitude, the peak deviation of the function from zero.
- f = the ordinary frequency, the number of oscillations (cycles) that occur each second of time.
- $\omega = 2\pi f$, the angular frequency, the rate of change of the function argument in units of radians per second
- φ = the phase, specifies (in radians) where in its cycle the oscillation is at t = 0.
- When φ is non-zero, the entire waveform appears to be shifted in time by the amount φ/ω seconds. A negative value represents a delay, and a positive value represents an advance.

2.2.2 Cosine Wave

A cosine wave is a signal waveform with a shape identical to that of a sine wave, except each point on the cosine wave occurs exactly 1/4 cycle earlier than the corresponding point on the sine wave. A cosine wave and its corresponding sine wave have the same frequency, but the cosine wave leads the sine wave by 90 degrees of phase.

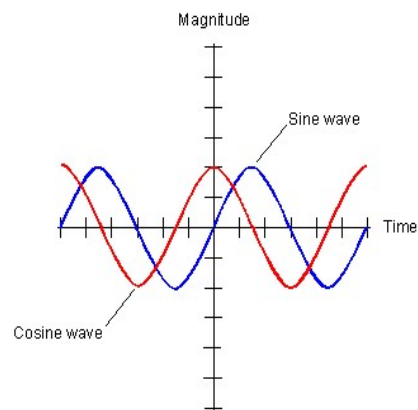


Figure 2.1: Cosine Wave

2.3 Sine and Cosine Wave *(Cont.)*

Creating a sine wave

```
1 x = 0:pi/100:2*pi;
2 y = sin(x);
3 plot (x, y)
```

The semicolon `[:]` in there means 'until'. For example, where the `x` stand for `(0,0.1,0.2,... 5)` the formula written `x=0:0.1:5`

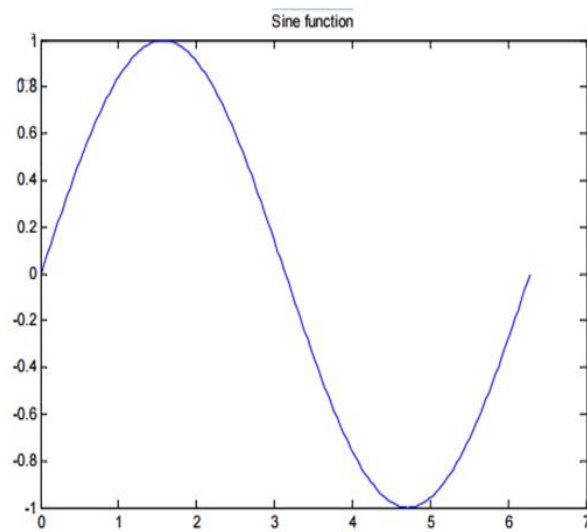


Figure 2.2: Sine Wave Example

2.4 Assignment

1. What is the output signal from Sine Wave below ?
2. What is the output signal from Cosine Wave below ?
3. Make your Conclusion about Lab Practice you did today!


```

Fs = 150; % Sampling frequency
t = 0:1/Fs:1; % Time vector of 1 second
f = 5; % Create a sine wave of f Hz.
x = sin(2*pi*t*f);
nfft = 1024; % Length of FFT
% Take fft, padding with zeros so that length(X)
is equal to nfft
X = fft(x,nfft);
% FFT is symmetric, throw away second half
X = X(1:nfft/2);
% Take the magnitude of fft of x
mx = abs(X);
% Frequency vector
f = (0:nfft/2-1)*Fs/nfft;
% Generate the plot, title and labels.
figure(1);
plot(t,x);
title('Sine Wave Signal');
xlabel('Time (s)');
ylabel('Amplitude');
figure(2);
plot(f,mx);
title('Power Spectrum of a Sine Wave');
xlabel('Frequency (Hz)');
ylabel('Power');

```

Figure 2.3: Code Assignment 1

```

Fs = 150; % Sampling frequency
t = 0:1/Fs:1; % Time vector of 1 second
f = 5; % Create a sine wave of f Hz.
x = cos(2*pi*t*f);
nfft = 1024; % Length of FFT
% Take fft, padding with zeros so that length(X) is
equal to nfft
X = fft(x,nfft);
% FFT is symmetric, throw away second half
X = X(1:nfft/2);
% Take the magnitude of fft of x
mx = abs(X);
% Frequency vector
f = (0:nfft/2-1)*Fs/nfft;
% Generate the plot, title and labels.
figure(1);
plot(t,x);
title('Sine Wave Signal');
xlabel('Time (s)');
ylabel('Amplitude');
figure(2);
plot(f,mx);
title('Power Spectrum of a Sine Wave');
xlabel('Frequency (Hz)');
ylabel('Power');

```

Figure 2.4: Code Assignment 2

2.5 Notes

Please write down your written task using Microsoft Word (format: DSP_LAB2_studentID.doc or docx) and the matlab code with the same name (format: DSP_LAB2_studentID.m). The tasks must be finished at lab (no take home lab task), late submission of works will not be marked.

Chapter 3

Square Wave Signal

3.1 Objectives

After a practicum assignment, it is expected that:

1. Students can Understand the meaning of Square Wave.
2. Students can implement the Square Wave formula into MATLAB Command.
3. Students can Represent the output Signal of Square Wave using MATLAB Program.

3.2 Theory

A square wave is a non-sinusoidal periodic waveform (which can be represented as an infinite summation of sinusoidal waves), in which the amplitude alternates at a steady frequency between fixed minimum and maximum values, with the same duration at minimum and maximum. The transition between minimum to maximum is instantaneous for an ideal square wave; this is not realizable in physical systems. Square waves are often encountered in electronics and signal processing. Its stochastic counterpart is a two-state trajectory. A similar but not necessarily symmetric wave, with arbitrary durations at minimum and maximum, is called a pulse wave (of which the square wave is a special case).

The square wave in mathematics has many definitions, which are equivalent except at the discontinuities:

It can be defined as simply the sign function of a periodic function, an example being a sinusoid:

$$x(t) = \text{sgn}(\sin[t])v(t) = \text{sgn}(\cos[t]) \quad (3.1)$$

which will be 1 when the sinusoid is positive, -1 when the sinusoid is negative, and 0 at the discontinuities. Any periodic function can substitute the sinusoid in this definition.

3.3 Square Wave Signal (Cont.)

To create square wave from sine wave, t must be define first.

```
1 t = 0:.1:.10;  
2 y = sin(t);  
3 plot (t, y)
```

The output signal is :

Now, you need to translate this formula to MATLAB Command :

$$y = \sin(t) + \sin\left(\frac{nt}{n}\right)$$

If we suppose that $n=3$, so

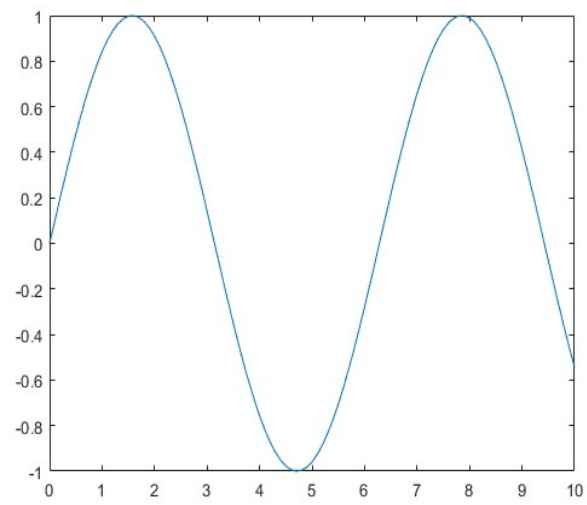


Figure 3.1: Square Wave Signal 1

```
1 y=sin(t) + sin(3*t)/3;  
2 plot(t,y);
```

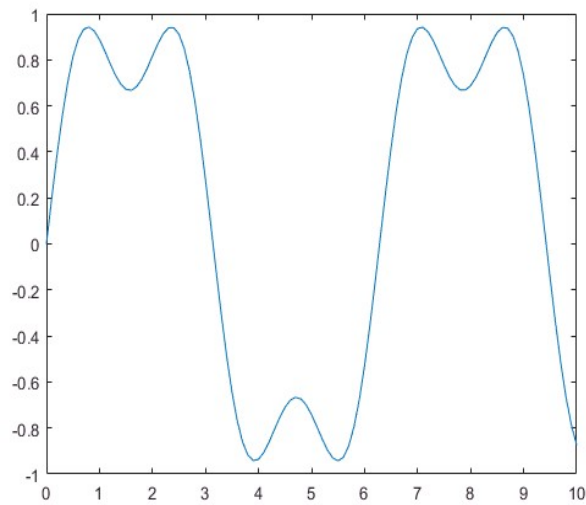


Figure 3.2: Square Wave Signal 2

3.4 Assignment

1. Convert these Square Wave (from Sine Wave) Formula into MATLAB Command!
 - $y = \sin(t) + \sin(\frac{5t}{5}) + \sin(\frac{7t}{7})$
 - $y = \sin(t) + \sin(\frac{3t}{3}) + \sin(\frac{5t}{5}) + \sin(\frac{7t}{7}) + \sin(\frac{9t}{9})$
2. What is the output signal from Cosine Wave below ?
 - See figure [3.3](#)
 - See figure [3.4](#)
3. Make your Conclusion about Lab Practice you did today!

```

Fs = 150; % Sampling frequency
t = 0:1/Fs:1; % Time vector of 1 second
f = 5; % Create a sine wave of f Hz.
x = square(2*pi*t*f);
nfft = 1024; % Length of FFT
% Take fft, padding with zeros so that length(X) is
equal to nfft
X = fft(x,nfft);
% FFT is symmetric, throw away second half
X = X(1:nfft/2);
% Take the magnitude of fft of x
mx = abs(X);
% Frequency vector
f = (0:nfft/2-1)*Fs/nfft;
% Generate the plot, title and labels.
figure(1);
plot(t,x);
title('Square Wave Signal');
xlabel('Time (s)');
ylabel('Amplitude');
figure(2);
plot(f,mx);
title('Power Spectrum of a Square Wave');
xlabel('Frequency (Hz)');
ylabel('Power');

```

Figure 3.3: Code Assignment 1

```

Fs = 150; % Sampling frequency
t = -0.5:1/Fs:0.5; % Time vector of 1 second
w = .2; % width of rectangle
x = rectpuls(t, w); % Generate Square Pulse
nfft = 512; % Length of FFT
% Take fft, padding with zeros so that length(X) is
equal to nfft
X = fft(x,nfft);
% FFT is symmetric, throw away second half
X = X(1:nfft/2);
% Take the magnitude of fft of x
mx = abs(X);
% Frequency vector
f = (0:nfft/2-1)*Fs/nfft;
% Generate the plot, title and labels.
figure(1);
plot(t,x);
title('Square Pulse Signal');
xlabel('Time (s)');
ylabel('Amplitude');
figure(2);
plot(f,mx);
title('Power Spectrum of a Square Pulse');
xlabel('Frequency (Hz)');
ylabel('Power');

```

Figure 3.4: Code Assignment 2

3.5 Notes

Please write down your written task using Microsoft Word (format: DSP_LAB3_studentID.doc or docx) and the matlab code with the same name (format: DSP_LAB3_studentID.m). The tasks must be finished at lab (no take home lab task), late submission of works will not be marked.

Chapter 4

Sawtooth Wave Signal

4.1 Objectives

After a practicum assignment, it is expected that:

1. Students can Understand the meaning of Sawtooth Wave.
2. Students can implement the Sawtooth Wave formula into MATLAB Command.
3. Students can Represent the output Signal of Sawtooth Wave using MATLAB Program.

4.2 Theory

The sawtooth wave (or saw wave) is a kind of non-sinusoidal waveform. It is so named based on its resemblance to the teeth of a plain-toothed saw with a zero rake angle. The convention is that a sawtooth wave ramps upward and then sharply drops. However, in a "reverse (or inverse) sawtooth wave", the wave ramps downward and then sharply rises. It can also be considered the extreme case of an asymmetric triangle wave.

The piecewise linear function

$$x(t) = t - [t] = t - \text{floor}(t) \quad (4.1)$$

or

$$x(t) = t(\text{mod}1) \quad (4.2)$$

T is an example of a sawtooth wave with period 1 based on the floor function of time t. A more general form is in the range of -1 to 1 and in the period a

$$x(t) = 2\left(\frac{t}{a} - \left[\frac{1}{2} + \frac{t}{a}\right]\right) = 2\left(\frac{t}{a} - \text{floor}\left(\frac{1}{2} + \frac{t}{a}\right)\right) \quad (4.3)$$

This sawtooth function has the same phase as the sine function. Another function in trigonometric terms with period p and amplitude a:

$$y(x) = \frac{-2}{\pi} \arctan\left(\cot\left(\frac{x\pi}{p}\right)\right) \quad (4.4)$$

While a square wave is constructed from only odd harmonics, a sawtooth wave's sound is harsh and clear and its spectrum contains both even and odd harmonics of the fundamental frequency. Because it contains all the integer harmonics, it is one of the best waveforms to use for subtractive synthesis of musical sounds, particularly bowed string instruments like violins and cellos, since the slip-stick behavior of the bow drives the strings with a sawtooth-like motion.

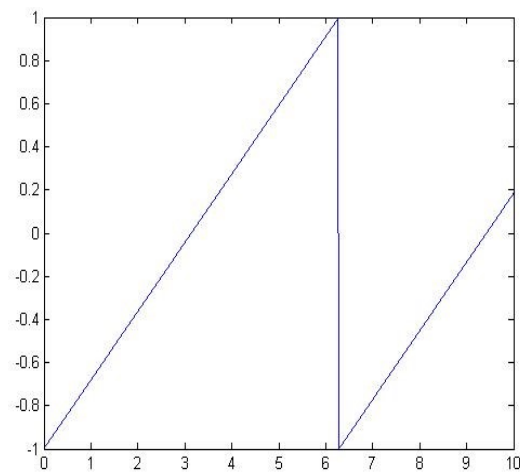


Figure 4.1: Sawtooth Signal

4.3 Sawtooth Wave Signal *(Cont.)*

To create Sawtooth Waveform, n must be define first.

```
1 n = (input signal value);
2 y = 0:0.002:n;
```

Suppose that n = 10, so;

```
1 n = 10;
2 t = 0:.0001:n;
3 y = sawtooth(t);
4 plot(t,y);
5 ylabel ('Amplitude');
6 xlabel ('Time Index');
7 TITLE ('Sawtooth waveform');
```

The output signal is :

Now, you need to translate this formula to MATLAB Command :

$$y = \sin(t) + \sin\left(\frac{nt}{n}\right)$$

If we suppose that n=3,so

```
1 y=sin(t) + sin(3*t)/3;
2 plot(t,y);
```

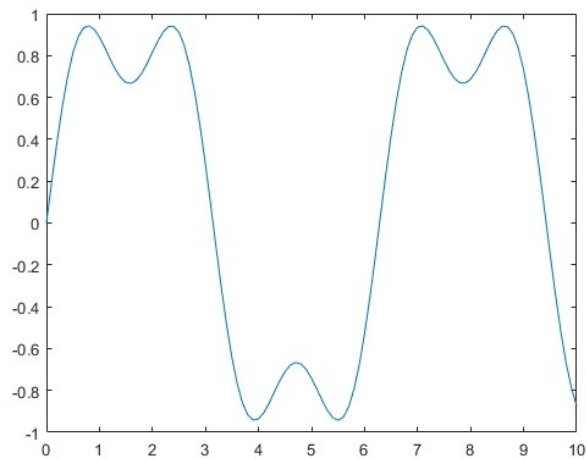


Figure 4.2: Square Wave Signal 2

4.4 Assignment

1. Input these n value of the Sawtooth Signal to MATLAB Command!
 - $n = 25$
 - $n = 50$
 - $n = 75$
 - $n = 100$
2. What is the output signal from Square Wave and Square Pulse below?
 - Modulate Sawtooth Signal using fm modulation. See figure [4.3](#)
 - Modulate Sawtooth Signal using fmod and fmdemod modulation. Seefigure [4.4](#)
3. Make your Conclusion about Lab Practice you did today!

```

fs=100000;
fm=2;
fc=1000;
t=0:1/fs:6/fm;
m=sawtooth(2*pi*fm*t);
plot(t,m);

```

Figure 4.3: Code Assignment 1

```

fs=3000;
fm=20;
fc=1000;
t=[0:fs-1]/fs;
x=sawtooth(2*pi*fm*t);
subplot(3,1,1);plot(x);
xlabel('main signal');
dev = 20;
y = fmmod(x,fc,fs,dev);
subplot(3,1,2);plot(y);
xlabel('modulated signal');
z = fmdemod(y,fc,fs,dev);
subplot(3,1,3);plot(z);
xlabel('demodulated signal');

```

Figure 4.4: Code Assignment 2

4.5 Notes

Please write down your written task using Microsoft Word (format: DSP_LAB4_studentID.doc or docx) and the matlab code with the same name (format: DSP_LAB4_studentID.m). The tasks must be finished at lab (no take home lab task), late submission of works will not be marked.

Chapter 5

Discrete Time Signal (*Pt.1*)

5.1 Objectives

After a practicum assignment, it is expected that:

1. Students can Understand the meaning of Discrete Time.
2. Students can implement the Discrete formula into MATLAB Command.
3. Students can Represent the output Signal of Discrete Time using MATLAB Program.

5.2 Theory

A discrete signal or discrete-time signal is a time series consisting of a sequence of quantities. In other words, it is a time series that is a function over a domain of integers.

Unlike a continuous-time signal, a discrete-time signal is not a function of a continuous argument; however, it may have been obtained by sampling from a continuous-time signal, and then arranging each value in the sequence is called a sample. When a discrete-time signal obtained by sampling a sequence corresponds to uniformly spaced times, it has an associated sampling rate; the sampling rate is not apparent in the data sequence, and so needs to be associated as a characteristic unit of the system.

5.3 Discrete Time Signal (*Pt.1–Cont.*)

To Create Signal Discrete Time by Unit Sample and Unit Step Formula

1. Unit Sample Signal

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & \neq 0 \end{cases}$$

The command in MATLAB Program, become;

```
1 n = -5:5;  
2 x = [n==0];  
3 stem (n,x)
```

The output signal is :

2. Unit Step Signal

$$v(n) = \begin{cases} 1, & n \geq 0 \\ 0, & < 0 \end{cases}$$

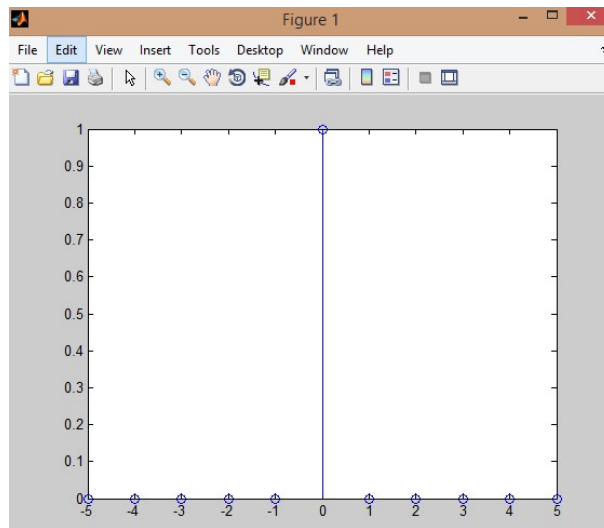


Figure 5.1: Discrete Time Signal 1

The command in MATLAB Program, become;

```
1 n = -5:5;  
2 x = [n>0];  
3 stem (n,x)
```

The output signal is :

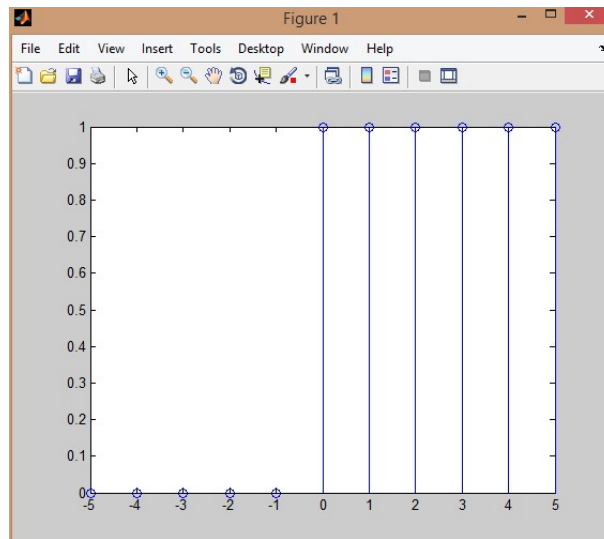


Figure 5.2: Discrete Time Signal 2

5.4 Assignment

1. Change the Discrete time signal sampling and step formula below into MATLAB command!

- $\delta(n - 5) = \begin{cases} 1, & n = 5 \\ 0, & n \neq 5 \end{cases}$

- $v(n - 5) = \begin{cases} 1, & n \geq 5 \\ 0, & n < 5 \end{cases}$

- $v_r(n - 1) = \begin{cases} 1, & n \geq 1 \\ 0, & n < 1 \end{cases}$

- $v_r(n - 2) = \begin{cases} n-2, & n \geq 2 \\ 0, & n < 2 \end{cases}$

2. What is the output signal from the MATLAB Command below?

- See figure [5.3](#)
- See figure [5.4](#)

3. Make your Conclusion about Lab Practice you did today!

```

n1 = 0:4;
x1 = [0 1 2 3 4];
n2 = -2:2;
x2 = [2 2 2 2 2];
n = min(min(n1),min(n2)):max(max(n1),max(n2)); % duration of y(n)
y1 = zeros(1,length(n)); y2 = y1; % initialization
y1(find((n>=min(n1))&(n<=max(n1))==1))=x1; % x1 with duration of y
y2(find((n>=min(n2))&(n<=max(n2))==1))=x2; % x2 with duration of y
y = y1+y2; % sequence addition
stem(n,y)

```

Figure 5.3: Code Assignment 1

```

%Discrete-time addition y(n)=x1(n)+x2(n)
n1 = 0:4;
x1 = [0 1 2 3 4];
subplot(3,1,1);stem(n1,x1); title('x_1(n) signal');
xlabel('n'); ylabel('x_1(n)');

n2 = -2:2;
x2 = [2 2 2 2 2];
subplot(3,1,2);stem(n2,x2); title('x_2(n) signal');
xlabel('n'); ylabel('x_2(n)');

n = min(min(n1),min(n2)):max(max(n1),max(n2)); % duration of y(n)
y1 = zeros(1,length(n)); y2 = y1; % initialization
y1(find((n>=min(n1))&(n<=max(n1))==1))=x1; % x1 with duration of y
y2(find((n>=min(n2))&(n<=max(n2))==1))=x2; % x2 with duration of y
y = y1+y2; % sequence addition
% addition
subplot(3,1,3); stem(n,y); title('Addition of the discrete-time signals');
xlabel('n'); ylabel('y(n)');

```

m-file

Figure 5.4: Code Assignment 2

5.5 Notes

Please write down your written task using Microsoft Word (format: DSP_LAB5_studentID.doc or docx) and the matlab code with the same name (format: DSP_LAB5_studentID.m). The tasks must be finished at lab (no take home lab task), late submission of works will not be marked.

Chapter 6

Discrete Time Signal (*Pt.2*)

6.1 Objectives

After a practicum assignment, it is expected that:

1. Students can Understand the meaning of Discrete Time.
2. Students can implement the Discrete formula into MATLAB Command.
3. Students can Represent the output Signal of Discrete Time using MATLAB Program.

6.2 Theory

Discrete time signals are “the signals or quantities that can be defined and represented at certain time instants of the sequence.” These are finite or countable sets of number sequences. They are also called digitalized signals. A discrete signal is shown in Figure 1.

One of the fundamental problems in signal processing is to obtain a permanent record of the signal itself. The problem with these analog recordings is that they are not abstract signals but a conversion of a physical phenomenon into another physical phenomenon: the temperature, for instance, is converted into the amount of ink on paper while the sound pressure wave is converted into the physical depth of the groove. The advent of electronics did not change the concept: an audio tape, for instance, is obtained by converting a pressure wave into an electrical current and then into a magnetic deflection. The fundamental consequence is that, for analog signals, a different signal processing system needs to be designed explicitly for each specific form of recording.

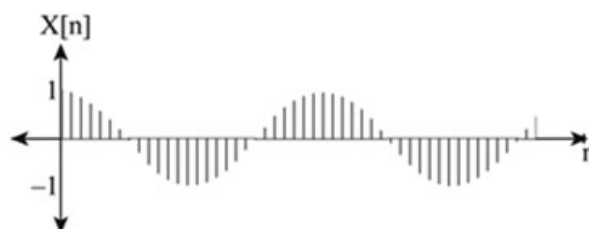


Figure 1

Figure 6.1: Discrete Time Signal 1

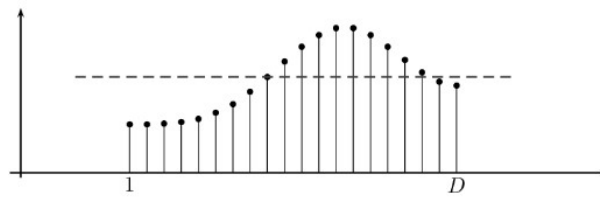


Figure 6.2: Discrete Time Signal 2

6.3 Discrete Time Signal *(Pt.2-Cont.)*

To Create Signal Discrete Time by Unit Complex-valued and Unit Sine and Cosine Formula

1. Unit Complex-valued Exponential Signal

$$x(n) = e^{(\varrho + j0)n}$$

The command in MATLAB Program, become;

```
1 n = -5:40;
2 x = (exp((3*4j)*n)) .* [n>=0];
3 y = real(x);
4 subplot(2,1,1);
5 stem(n,y)
6 z = imag(x);
7 subplot(2,1,2);
8 stem(n,z)
```

The output signal is :

$$x(n) = 4 * \cos(0.1\pi n + \frac{\pi}{3})$$

The command in MATLAB Program, become;

```
1 n = -5:40;
2 x = 4*cos(0.1*pi*n+pi/3);
3 stem(n,x)
```

The output signal of the command is :

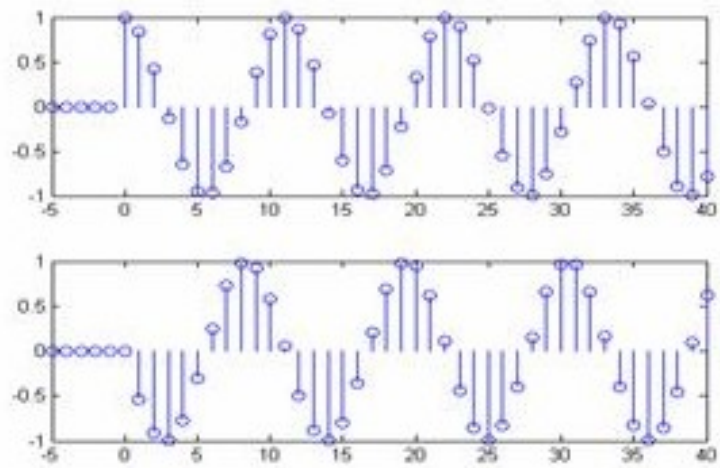


Figure 6.3: Discrete Time Signal 1

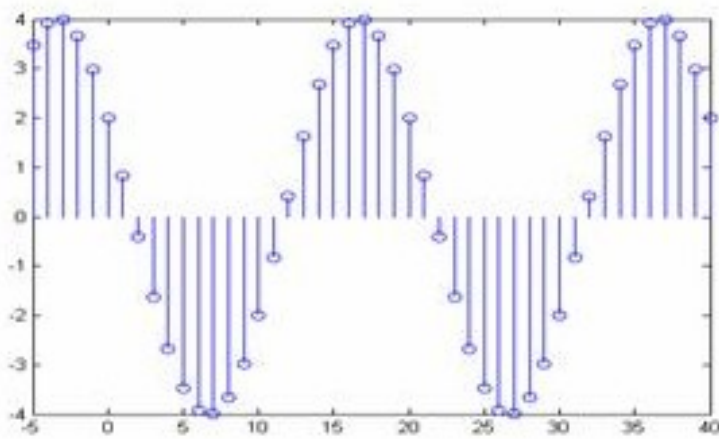


Figure 6.4: Discrete Time Signal 2

6.4 Assignment

1. Change the Discrete time signal Sine and Cosine formula below into MATLAB command!

- $x(n) = 5\cos(0.1\pi n + \frac{\pi}{4}) + 4\sin(0.3\pi n + \frac{\pi}{4})$
- $x(n) = 2\sin(0.3\pi n + \frac{\pi}{3}) + 4\sin(0.2\pi n + \frac{\pi}{4})$
- $x(n) = 3 * \cos(0.5\pi n + \frac{\pi}{3}) + 4\sin(0.3\pi n + \frac{\pi}{4})$
- $x(n) = 4 * \sin(0.4\pi n + \frac{\pi}{3}) + 4\sin(0.3\pi n + \frac{\pi}{4})$

2. What is the output signal from the MATLAB Command below?

- See figure [6.5](#)
- See figure [6.6](#)

3. Make your Conclusion about Lab Practice you did today!

```

%Shifting a non-function Discrete-time
signal
n = 0:8;
x = [0 1 5 2 1 3 6 4 5];
subplot(2,1,1);stem(n,x); title('
signal');
xlabel('n'); ylabel('x(n)');

m=n+2; y=x;
subplot(2,1,2);stem(m,y); title('y(n)=x(n-
2) signal');
xlabel('n'); ylabel('y(n)');

```

m-file

Figure 6.5: Code Assignment 1

```

%Folding a Discrete-time signal
n = 0:8;
x = [0 0 1 2 3 4 5 4 3];
subplot(2,1,1);stem(n,x); title('x(n) signal');
xlabel('n'); ylabel('x(n)');

m=-fliplr(n); y=fliplr(x);
subplot(2,1,2);stem(m,y); title('y(n)=x(-n)
signal');
xlabel('n'); ylabel('y(n)');

```

m-file

Figure 6.6: Code Assignment 2

6.5 Notes

Please write down your written task using Microsoft Word (format: DSP_LAB6_studentID.doc or docx) and the matlab code with the same name (format: DSP_LAB6_studentID.m). The tasks must be finished at lab (no take home lab task), late submission of works will not be marked.